Topic Modeling & Bibliometric Analysis Pipeline

This repository provides an end-to-end R-based workflow for text preprocessing, Latent Dirichlet Allocation (LDA) topic modeling, evaluation, visualization, and bibliometric exploration using `bibliometrix`. It is designed for researchers analyzing scientific literature datasets (e.g., Scopus or Web of Science exports).

Project Structure

```
README.md
/data/            # raw input files (e.g., scopus_export.csv)
/scripts/         # main R scripts
/output/          # generated CSVs, plots, and models
/notebooks/       # optional RMarkdown notebooks
/LICENSE
```

Workflow Overview

1. Load & Import Data

Uses: `readtext`, `quanteda`, `textstem`, `tm`, `dplyr`
Input expected: a CSV or text file containing Title, Abstract, and Author Keywords columns.

```
x <- readtext(file.choose())
x$data <- paste(x$Title, x$Abstract, x$`Author Keywords`)
```

2. Text Cleaning & Normalization

- Convert text to lowercase.
- Lemmatize using `textstem::lemmatize_strings()`.
- Remove numbers, punctuation, and English stopwords using `tm`.

3. Tokenization & Stopword Filtering

- Tokenize text via `quanteda::tokens()`.
- Remove stopwords and short tokens (`min_nchar = 2`).
- Apply a large custom stopword list to remove domain-generic words (e.g., study, paper, result).

4. Phrase Merging

Combine frequent multi-word expressions for improved topic coherence:

```
t1 <- gsub("\\b(social|medium)\\b", "social_media", t, perl = TRUE)
t1 <- gsub("\\b(machine|learn)\\b", "machine_learning", t1, perl = TRUE)
```

5. Create Document-Feature Matrix (DFM)

```
d1 <- quanteda::dfm(t1)
topfeatures(d1, 100)
```

Topic Modeling (LDA)

Fit Multiple Models

Fit LDA models for k = 2 to 15 topics using Gibbs sampling:

```
lda5 <- LDA(d2, k = 5, method = "Gibbs", control = list(seed = 1234))
```

Evaluate Perplexity

```
perplexity(lda5, newdata = d2)
```

Determine Optimal Number of Topics

Using `ldatuning::FindTopicsNumber()`:

```
result <- FindTopicsNumber(
  dtm,
  topics = seq(2, 15, 1),
  metrics = c("Griffiths2004", "CaoJuan2009", "Arun2010", "Deveaud2014"),
  method = "Gibbs",
  control = list(seed = 1234)
)
FindTopicsNumber_plot(result)
```

Visualization & Labeling

Top Terms per Topic

```
library(tidytext)
library(ggplot2)

terms <- tidy(lda9, matrix = "beta") %>%
  group_by(topic) %>%
  slice_max(beta, n = 10)
```

Plot

Each facet displays top 10 terms per topic with their β-values.

```
ggplot(terms, aes(beta, term, fill = factor(topic))) +
  geom_col() + facet_wrap(~ topic)
```

Topic Relationship Analysis

Topic Similarity Heatmap

Compute cosine similarity between topics:

```
similarity_matrix <- text2vec::sim2(posterior(k)$terms, method = "cosine")
corrplot(similarity_matrix, method = "color")
```

Topic Network Graph

```
g <- igraph::graph_from_adjacency_matrix(similarity_matrix, mode =
"undirected", weighted = TRUE)
plot(g)
```

Intertopic Distance Map (MDS)

```
mds <- cmdscale(1 - similarity_matrix, k = 2)
ggplot(...)
```

Document–Topic Exploration

Extract most representative documents per topic to interpret thematic meaning.

```
doc_topics <- posterior(lda9)$topics
top_docs[[topic]] <- order(doc_topics[, topic], decreasing = TRUE)[1:3]
```

Bibliometric Analysis

Launch Biblioshiny for interactive bibliometric exploration:

```
library(bibliometrix)
biblioshiny()
```

Installation

Install required R packages:

```
install.packages(c(
  'readtext','quanteda','textstem','tm','topicmodels','ldatuning',
  'tidytext','ggplot2','text2vec','igraph','corrplot','bibliometrix'
))
```

Outputs

File	Description
`topics.csv`	Top words per topic
`Top_10_Terms_Per_Topic.csv`	β-values for top terms
`griffith2004.csv`	Topic evaluation metrics
`topic_network.png`	Network visualization of topic similarity
`mds_intertopic_map.png`	Intertopic distance visualization

Notes & Recommendations

- Validate topic interpretability using top terms and representative documents.
- Use `dfm_trim()` to reduce sparsity and speed up modeling.
- Save models via `saveRDS()` for reproducibility.
- Ensure consistent random seeds (`seed = 1234`).

Future Extensions

- Add coherence scoring (via `textmineR` or `ldatuning`).
- Build a Shiny dashboard for interactive topic inspection.
- Integrate dynamic topic modeling (DTM) for temporal trend analysis.

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